MA.8.GR.1.2

Benchmark

MA.8.GR.1.2 **Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.**

Example: The distance between $(-2, 7)$ and $(0, 6)$ can be found by creating a right triangle with the vertex of the right angle at the point $(-2, 6)$. This gives a height of the right triangle as 1 unit and a base of 2 units. Then using the Pythagorean Theorem the distance can be determined from the equation $1^2 + 2^2 = c^2$, which is equivalent to $5 = c^2$. So, the distance is $\sqrt{5}$ units.

Benchmark Clarifications:
Clarification 1: Instruction includes making connections between distance on the coordinate plane and right triangles.
Clarification 2: Within this benchmark, the expectation is to memorize the Pythagorean Theorem. It is not the expectation to use the distance formula.
Clarification 3: Radicands are limited to whole numbers up to 225.

Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.1, MA.8.NSO.1.2, MA.8.NSO.1.7
- MA.8.AR.2.3

Terms from the K-12 Glossary

- Coordinate
- Coordinate Plane

Vertical Alignment

Previous Benchmarks

- MA.6.GR.1.2, MA.6.GR.1.3

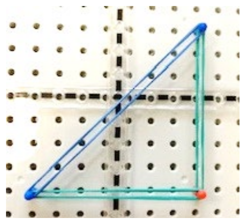
Next Benchmarks

- MA.912.GR.3.2, MA.912.GR.3.3, MA.912.GR.3.4
- MA.912.GR.7.2
- MA.912.T.1.2

Purpose and Instructional Strategies

In grade 6, students used their understanding of the coordinate plane to plot rational-number ordered pairs in all four quadrants and on both axes, and they found the distances between ordered pairs with the same x-coordinate or the same y-coordinate represented on the coordinate plane. In grade 8, students find the distance between two points using the Pythagorean Theorem. In Geometry, students will use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles or quadrilaterals. Additionally, students will extend this understanding to using coordinate geometry and trigonometry to solve mathematical and real-world problems involving lines, circles, triangles, quadrilaterals and finding the perimeter or area of polygons.

- Instruction includes creating a right triangle from two given points and then using the Pythagorean Theorem to find the distance between the two given points. This work can be started by using Geoboards to see the triangle that is formed within the coordinate plane. Students can show how to make a right triangle using vertical and horizontal lines. From there they can build the area models of the Pythagorean Theorem to support understanding.



- Students should be given multiple opportunities to see the importance of using the coordinate plane to find the distance between two points.
- Instruction includes providing students with a structure to support the organization of their work since using the Pythagorean Theorem may require multiple steps. Provide students with resources, including the coordinate plane and graph paper, as a way to plan out their work.

Common Misconceptions or Errors

- Students may have the misconception that the Pythagorean Theorem will apply to any triangle.
- Students may invert the x- and y-value of the point.
- When finding distances that cross over an axis students may incorrectly use operations with integers.
    - For example, if given the points $(-2, 0)$ and $(3, 0)$, students may calculate the distance as 1 unit instead of 5 units.

Strategies to Support Tiered Instruction

- Instruction includes the use of geometric software to explore the Pythagorean Theorem on obtuse, acute and right triangles.
- Instruction includes students adding the absolute value of two x-coordinates or two y-coordinates when the given points cross over an axis.
    - For example, if the given points are $(-4, 8)$ and $(7, 8)$, students will add the absolute value of -4 and 7.

$$|-4| + |7| = 11$$

- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions.
    - What do you know from the problem?
    - What is the problem asking you to find?
    - Can you create a visual model to help you understand or see patterns in your problem?
- Instruction includes labeling the x- and y-value of a coordinate point before graphing to reinforce the process of graphing x- and y-values.

- Instruction includes laying trace paper on top of a coordinate plane, tracing the points, drawing a number line through the two points, and counting the space between the points to find the distance.
- Teacher co-creates an anchor chart while students create a similar own graphic organizer to include key features of a coordinate plane. Features include the x-axis, y-axis, origin, quadrants, numbered scales and an ordered pair.
- Instruction includes the use of a three-read strategy. Students read the problem three times, each with a different purpose (laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful).
  - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
  - Second, read the problem with the purpose of answering the question: What are we trying to find out?
  - Third, read the problem with the purpose of answering the question: What information is important in the problem?

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.7.1)

Pineridge Middle School was given a grant from Home Helper Depot to create a triangular garden along a wall of the cafeteria for fresh vegetables. The length of the hypotenuse and the sides are being determined to see if it will fit in the space. On the model for the garden, the designer started by plotting the points $(2, 2)$ and $(6, 5)$ on a coordinate plane and connected the points with a line. She needs to complete the triangular model and determine all three side lengths.

Part A. Using a coordinate grid, complete the designer's drawing.
Part B. Calculate the side lengths of the triangular garden on the model.
Part C. What would be appropriate lengths for a triangular garden if the length of one side of the building is 20 feet? Use your model to help determine the side lengths.

Instructional Items

Instructional Item 1

On a coordinate plane, plot the points $(-3, 4)$ and $(0, -3)$. Using the Pythagorean Theorem, determine the distance between the two points.

Instructional Item 2

Using the Pythagorean Theorem, determine the distance from point $(8, -6)$ to the origin.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*